

E-Commerce Platform Draft Requirements

The goal is to build an online store that lets customers register, browse items, add them to a shopping cart, and pay securely. This document lists early requirements and notes from the product team.

1. User Accounts and Login

Users should be able to create an account using their email or sign in with Google. Passwords must be stored securely, and password reset should be available through an email link.

If a user enters the wrong password three times, the system should temporarily lock the account for 15 minutes. There should be a “remember me” option on login.

When logged in, customers can view their order history and saved addresses. Guest checkout should also be supported, but returning users will benefit from saved preferences.

2. Product Catalog and Search

The site must support multiple product categories and allow filtering by price, size, brand, or availability. Users should be able to search products by keyword or partial name.

Each product will have a detail page with an image gallery, description, price, and customer reviews. The product page should also show stock availability and estimated delivery time.

Admins can update product details through the dashboard, including setting sale prices and marking items as “featured.”

If a product is out of stock, customers should be able to subscribe for an email notification when it becomes available again.

3. Shopping Cart and Wishlist

Items can be added to a shopping cart from any product page. The cart should persist between sessions if the user is logged in. Users can change quantities or remove items before checkout.

There should also be an option to “save for later” — stored as a wishlist linked to the user’s account.

Cart totals should update automatically when quantities change. If a promo code is applied, it should immediately update the total cost.

4. Checkout and Payment

Checkout should be a multi-step process: address, shipping method, payment, and confirmation. Guest checkout should prompt users for an email to send the receipt.

We'll integrate with Stripe and PayPal for payments. All payment processing must be secure and compliant with PCI DSS standards.

The system must validate credit card info before submission. After successful payment, users should receive a confirmation email with an order summary.

If payment fails, the order should not be created. Errors should be clear and actionable (e.g., "Card expired").

5. Order Management

After checkout, an order should be created in the system with a unique ID. Admins can update the status (Pending, Shipped, Delivered). Customers should be able to track their order from their profile.

An automatic email should be sent when the order status changes. There should also be a refund process — initiated by the admin, notifying the user.

6. Admin Dashboard

The admin dashboard should show total sales, number of orders, and top-selling products. Admins can manage inventory, edit product info, and view user activity logs.

There should be role-based access control:

- Admin: full permissions
- Manager: product/order access
- Support: view-only access

Reports should be exportable to CSV or Excel.

7. Performance and Security Notes

The site must load in under 3 seconds on a stable connection. All pages must use HTTPS. Sensitive data (passwords, credit cards) should be encrypted in transit and never logged.

8. Notifications and Emails

SendGrid will be used for sending transactional emails (order confirmation, password reset, etc.). All emails should include company branding and links to order tracking.

9. Future Enhancements (Ideas)

- Product recommendations based on browsing history.
- Loyalty program for frequent customers.
- Multi-language support (English, Spanish).